MECH191 — Metallurgy

Week 7

Mechanical Testing Methods

October 5, 2026 · 5:00 – 7:30 PM · Kai Santos

Class Agenda

01

Tensile & Hardness Testing

What each test measures, how to read results, and which method fits which material

02

Impact & Fatigue Testing

Charpy impact, the ductile-brittle transition, fatigue mechanisms, and the S-N curve

03

Reading Mill Test Reports

Traceability, acceptance criteria, common flags, and the technician's responsibility

Metallography Lab — Module 6

Module 6 · Week 7

Mechanical Testing

What to do this week:

Complete ASTM E112 grain size intercept count on Specimen A (1018 steel). Measure and calculate HV from indent on both specimens.

Steps:

1 Observe — complete feature table
2 Sketch — label magnification & specimen
3 Answer — 4 short questions

Specimens to Check Out

- Specimen A — 1018 steel (Vickers indent)
- Specimen B — 1045 steel (Vickers indent)

How to check out the kit:

1 See instructor after class today
2 Sign out the specimen tray
3 Use the EXAMET-5 at your convenience
4 Return kit before leaving class
5 Submit completed module with your portfolio

MECH 191 — Course & Unit Roadmap

Unit 1 Materials Fundamentals	Unit 2 Steel & Alloys	Unit 3 Testing & Quality	Unit 4 Manufacturing	Unit 5 Joining & Failure
Wk 1 Aug 24 Atomic Structure & Bonding	**Wk 4** Sep 14 Carbon & Alloy Steels	**Wk 7 ← here** Oct 5 Mechanical Testing	**Wk 10** Oct 26 Casting	**Wk 14** Nov 23 Welding Processes
Wk 2 Aug 31 Mechanical Properties	**Wk 5** Sep 21 Stainless & Cast Irons	**Wk 8** Oct 12 NDT Methods	**Wk 11** Nov 2 Forming	**Wk 15** Nov 30 Weld Metallurgy & Failure
	Wk 6 Sep 28 Nonferrous & Phase Diagrams	**Wk 9** Oct 19 Quality Standards	**Wk 12** Nov 9 Machining	**Wk 16** Dec 7 Corrosion & Wrap-up
			Wk 13 Nov 16 Powder Metallurgy	

Week 17 · FINAL EXAM

Week 7 — At a Glance

Topics

- Tensile testing — stress-strain curve, yield strength, UTS, elongation at fracture
- The 0.2% offset method for materials without a clear yield point
- Hardness testing — Rockwell (HRC, HRB), Brinell (HB), and Vickers (HV): methods and when each is used
- Charpy impact testing — notched specimens, energy absorption, and the ductile-brittle transition temperature
- Fatigue — cyclic loading, crack initiation at stress concentrations, the S-N curve, and the steel fatigue limit
- Reading a mill test report (MTR) — heat numbers, composition, mechanical test results, and traceability

Resources

Reference Material

Moniz Ch. 3
Mechanical Testing

Kalpakjian Ch. 2
Fundamentals of Mechanical Behavior

Visualizers

Stress-Strain Curve (Interactive)

Reference Pages

Hardness Testing Scales
Impact & Fatigue Testing

Core Questions

Try to answer these before we dive in — we'll revisit them with answers at the end of class.

1. What does a tensile test actually measure and what do the results tell you about a material?
2. What is the difference between Rockwell and Brinell hardness, and when is each used?
3. Why is impact testing important and what is the ductile-brittle transition?
4. What is fatigue and why can a metal fail at stresses well below its yield strength?
5. How do you read a mill test report — and what are the key items a technician must verify?

Tensile Testing — The Stress-Strain Curve

What the Test Measures

- Pulls a standardised specimen apart at a controlled rate while measuring force and extension
- Force and extension are converted to engineering stress ($\sigma = F/A_0$) and engineering strain ($\varepsilon = \Delta L/L_0$)
- Results are plotted as a stress-strain curve — each point tells a different story about the material
- The 0.2% offset method is used when no clear yield point exists — draw a line parallel to the elastic slope from 0.2% strain; the intersection with the curve is the reported yield strength
- A single tensile test returns yield strength, UTS, elongation, and reduction in area — more data per test than any other routine test

Key Values from the Curve

Proportional limit:

Stress up to which σ–ε is linear — the elastic region

Yield strength (Sy):

Stress at onset of permanent plastic deformation — a primary design parameter

Ultimate tensile strength (UTS):

Maximum stress on the curve; necking begins at this point

Elongation at fracture:

Total permanent strain at fracture (%) — direct measure of ductility

Reduction in area:

% decrease in cross-section at fracture — especially useful for materials that neck heavily

Curve shape:

Long plastic plateau = tough and ductile; steep then abrupt fracture = brittle; no clear yield point = typical aluminum

Hardness Testing — Three Methods, Three Applications

Method Comparison

Method	Indenter	Load & Readout	Best For	Typical Scale
Rockwell	Diamond cone (HRC) Steel ball (HRB)	Minor + major load; depth readout instantly	Production inspection; finished parts	HRC 20–70 (hard steels) HRB 0–100 (soft metals)
Brinell	Hardened ball (10 mm, W/C)	High load; measure indent diameter optically	Castings, forgings, coarse-grained materials	HB / BHN up to ~650; approximate UTS from HB
Vickers	Square pyramid diamond (same for all)	Variable low loads; measure diagonals optically	Thin sections, coatings, precision & microhardness	HV 1–3000; converts across HRC/HB

Conversion Caution

Approximate conversion tables between HRC, HB, and HV exist — and approximate UTS can be estimated from HB (UTS ≈ 3.45 × HB in MPa for carbon steels). These are approximations: use with caution on alloy steels and non-ferrous materials where the relationship differs.

Technician Rule of Thumb

Use Rockwell on the production floor for quick pass/fail checks. Use Brinell on rough or coarse-grained material where a small indentation would land in a soft spot. Use Vickers when measuring case depth, thin-wall sections, or individual microstructural phases.

Impact Testing & the Ductile-Brittle Transition

The Charpy Impact Test

- A notched rectangular specimen is supported at both ends; a swinging pendulum strikes from behind at the notch
- Energy absorbed = $mgh_1 - mgh_2$ (the difference between swing height before and after fracture), reported in Joules or ft-lbf
- The notch simulates a real stress concentration — a crack tip, undercut weld, sharp corner, or inclusion
- Tough materials absorb high energy and show large lateral expansion and shear fracture surfaces; brittle materials absorb little energy and show bright, flat, cleavage fracture surfaces
- Izod test: specimen clamped vertically and struck at the top — less common in modern standards than Charpy

Ductile-Brittle Transition (DBTT)

Upper shelf:

High absorbed energy — fully ductile fracture. Material behaves toughly above the transition temperature.

Lower shelf:

Low absorbed energy — fully brittle fracture. Even a tough-looking material becomes dangerous below this temperature.

Transition temperature:

Defined at 50% of upper shelf energy (or 50% brittle fracture appearance). A material can pass room-temperature testing but be catastrophically brittle at service temperature.

BCC steels:

Carbon and low-alloy steels show a sharp DBTT — critical for arctic pipelines, offshore structures, and pressure vessels in cold service.

FCC metals:

Austenitic stainless, aluminum, copper remain tough at cryogenic temperatures — no DBTT. Essential for LNG and cryogenic applications.

Fatigue — Cyclic Loading and the S-N Curve

How Fatigue Works

- Fatigue is progressive crack initiation and growth under cyclic loading — even when applied stress is well below yield strength
- Each load cycle causes tiny damage at a stress concentration; damage accumulates invisibly over thousands or millions of cycles
- No plastic deformation occurs until final fracture — a fatigue failure gives NO visible warning beforehand
- Crack initiation almost always occurs at a stress concentration: surface scratch, tool mark, sharp corner, weld toe, porosity, inclusion, or notch
- Compressive residual stress (e.g. from shot peening) delays crack initiation and improves fatigue life; tensile residual stress (e.g. from welding) accelerates it

The S-N Curve (Wöhler Curve)

What it plots:

Stress amplitude (S) vs. number of cycles to failure (N) on a log scale — a downward-sloping line: lower stress survives more cycles.

Fatigue limit (steel):

Typically 40–50% of UTS for carbon steels. Below this stress amplitude, a steel can theoretically survive an indefinite number of cycles — a genuine design limit.

Aluminum alloys:

Have NO true fatigue limit — the S-N curve continues to slope downward. Fatigue life must be specified for a finite number of cycles (e.g. 10^7 cycles).

Fatigue ratio:

Fatigue limit ÷ UTS — useful for comparing materials; typically 0.4–0.5 for steels, 0.3–0.4 for aluminum alloys.

Design implication:

Surface finish, notches, and residual stresses affect where the S-N curve sits — a polished surface in compression performs far better than a notched surface in tension.

Reading a Mill Test Report (MTR)

The MTR is the chain of traceability between raw material and finished component — without it, there is no way to verify the material meets the specification.

01 Heat / Lot Number

Must match the number stamped or stencilled on the physical material. Without this match, the MTR does not certify the material in your hand.

02 Material Grade & Specification

Confirm it matches what was ordered — ASTM A36, API 5L Grade B, etc. A grade substitution not approved by engineering is a nonconformance.

03 Chemical Composition

Compare each element against the specification limits — carbon, manganese, phosphorus, and sulfur always; alloying elements as required by the grade.

04 Mechanical Test Results

Yield strength, UTS, and elongation at minimum; impact values if required. Compare each against the minimum (or maximum) acceptance criteria.

05 Test Standard & Certifying Signature

Tests must be performed to the correct standard (e.g. ASTM A370). The document must be signed and traceable to an accredited testing facility.

Technician's Perspective — When Results Have Real Consequences

Red Flag: MTR heat number mismatch

The heat number on the MTR does not match the number stencilled on the pipe. Stop — do not use the material. Tag it as quarantined, issue a nonconformance report, and notify the quality engineer. A material whose identity cannot be confirmed cannot be installed in a code-governed system.

Caution: Impact values just at minimum

The Charpy values in the MTR meet the specification — but they are at the minimum allowable limit on a project with a low minimum service temperature. Flag it to the engineer before installation. Minimum-conforming is not always fit-for-purpose when temperatures drop below the tested range.

Good Practice: Hardness survey after welding

After welding P91 chrome-moly pipe, a Vickers microhardness survey is run across the weld, HAZ, and base metal. Hard spots above the limit indicate insufficient post-weld heat treatment — the weld must be heat-treated again before pressure testing. The test caught a problem before the system went into service.

Core Questions — Answers

What does a tensile test measure and what do the results tell you?

Pulls a specimen apart while recording force and extension → stress-strain curve. Gives yield strength (permanent deformation onset), UTS (maximum stress), and elongation (ductility). Curve shape reveals whether the material is ductile or brittle.

Rockwell vs. Brinell hardness — what's the difference and when is each used?

Rockwell measures penetration depth directly — fast, suits production inspection of finished parts (HRC for hard steels, HRB for softer metals). Brinell measures a large indentation optically — averages out microstructural variation, suits castings and forgings.

Why is impact testing important and what is the ductile-brittle transition?

Tensile tests apply load slowly; service impact is sudden. BCC steels have a DBTT — above it they absorb high energy (ductile); below it they fracture with little warning (brittle). Charpy tests verify adequate toughness at minimum service temperature.

What is fatigue and why can a metal fail below yield strength?

Cyclic loading initiates a crack at a stress concentration, growing invisibly with each cycle until sudden fracture. No visible warning because yield is never reached. The S-N curve gives safe design stress; steels have a fatigue limit; aluminum alloys do not.

How do you read a mill test report?

Check: ① heat number matches material markings; ② grade/spec matches order; ③ chemical composition within spec limits; ④ mechanical results (yield, UTS, elongation, impact if required) meet minimums; ⑤ test standard and certifying signature are present.

Key Takeaways

1

Mechanical testing translates microstructure into numbers — tensile, hardness, impact, and fatigue tests each capture a different aspect of material behavior, and a technician who can read and interpret the results can make informed decisions about material acceptance, fitness for service, and failure investigation.

2

Hardness is the fastest and most practical field test, but must be matched to the right scale and method for the material — Rockwell for production inspection of finished parts, Brinell for castings and forgings where microstructural variation matters, Vickers for precision work and thin sections.

3

The mill test report is the documentary backbone of material traceability — verifying that the heat number, grade, composition, and mechanical results all match the specification is a fundamental quality control responsibility that every technician must take seriously before any material goes into service.

Resources & What's Next

Week 7 Resources

Moniz Ch. 3
Mechanical Testing — primary reading

Kalpakjian Ch. 2
Fundamentals of Mechanical Behavior of Materials

Stress-Strain Visualizer
Interactive — plot your own material parameters

Hardness Scales Reference
HRC / HRB / HB / HV conversion chart

Impact & Fatigue Reference
Charpy specimen geometry, S-N curve examples

Week 8 — NDT Methods

October 12, 2026

- Visual testing (VT) — the first step before any other method
- Dye penetrant (PT) — finding surface-breaking defects in any non-porous material
- Magnetic particle (MT) — near-surface defects in ferromagnetic materials
- Ultrasonic (UT) — internal defect detection and wall thickness measurement
- Radiographic (RT) — volumetric imaging and the radiation safety requirements that come with it